

1 Semantic Seed Selection

User-related facts

PersonaMem-v2

User profiles



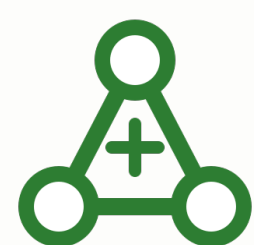
Non-user facts

FanOutQA, MuSiQue, QACC, HoH, AmbigQA



2 Semantic Variants Creation

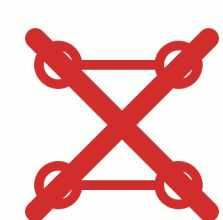
Generate relation-preserving valid traceable semantic variants.



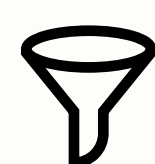
Complementary
mutually compatible
evidentiary variants



Nuanced
subtly distinguishable
semantic variants



Contradictory
mutually exclusive
applicable variants



Case-level Filter

- ✓ Remove invalid or duplicated variants
- ✓ Enforce compatibility relation fidelity
- ✓ Preserve source traceability

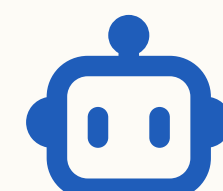
3 Session Construction

Embed fact variants into task-oriented multi-turn user-assistant sessions.



I'm updating my weekly meal plan...

I'll separate the weekday plan from Sunday dinner...



...



Session-level Filter

- ✓ Preserve fact inferability
- ✓ Ensure interaction naturalness
- ✓ Preserve relation consistency

4 Evaluation Instance Construction

Convert each case into evaluation instances.

Question (non-user)



Who plays Timothy McGee's wife on NCIS?
Correct: Margo Harshman.
Incorrect: Sean Murray.

Task (user-related)



Pick one item for how I save travel memories...
Correct: Album refill.
Incorrect: Travel journal.



Instance-level Filter

- ✓ Ensure target dependence
- ✓ Enforce relation non-disclosure
- ✓ Verify query-answer consistency

5 User-History Assembly

Build coherent time-ordered user histories.



Personal trajectory
User-related sessions are organized to form a coherent long-term personal history.



Context-Compatible Integration

Insert user-unrelated sessions where topic and timing remain compatible.



Final Histories
Interleave both memory types into realistic chronological histories.

Data Composition: **One user-history** maps to a set of evaluation instances.

A. History (time-ordered)

Interleaved sessions (5-10 turns each)

Session 215 (non-user)

User: I need help tightening a comic con panel blurb: "Oliver Stark, best known for playing Evan 'Buck' Buckley..."
Agent: Join Oliver Stark, best known for playing Evan "Buck" Buckley, for a live conversation...

Session 216 (non-user)

User: Draft: "Oliver Stark, who is Buck in 9-1..."
Agent: 9-1-1 star Oliver Stark, who plays Buck, chats about the show in a new Q&A...

Other sessions (user related & unrelated)

B. Queries

Only the history and the query are provided to agent.

Query

One quick check: who plays Buck on 9-1-1 again?

Instruction to the agent

Answer the user based on the given conversation history.

C. Answers (for judging)

Used only for automatic evaluation.

Correct Answers

Oliver Stark plays Buck on 9-1-1

Incorrect Answers

- Ryan Guzman plays Buck on 9-1-1
- Evan "Buck" Buckley is the actor who plays Buck on 9-1-1
- It's Kenneth Choi, who plays Buck on 9-1-1

D. Hidden Metadata (not shown to agent)

For automatic evaluation and analysis only.

- Source: non-user
- Case ID: unrelated-comp-qacc-00048
- Instance ID: unrelated-30adf5433abb
- Relation: Complementary
- Sub-type: any_one
- Facts:
 - Oliver Stark is best known for playing Evan "Buck" Buckley.
 - "9-1-1" star Oliver Stark is described as the actor who plays Buck.
- Session IDs:
 - unrelated-persona_0-comp-qacc-00048-s1
 - unrelated-persona_0-comp-qacc-00048-s2